

**Table 6: List of all tested rules of Google’s Car App Quality Guidelines in AUTOGUARD. The specific rules mentioned in this paper have been bolded.**

| - Categories          | Criteria                    | App                                                               | ID & Description                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|-----------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Functionality | Permitted Categories        | Media,Navigation,POI, IOT,Messaging,Video, Games,Browsers,Weather | <b>PC-1: The app must work as expected and not include features outside the app types.</b>                                                                                                                                                                                                                                                                            |
|                       | Expected Performance        | Media,Navigation,POI, IOT,Video,Games, Browsers,Weather           | <b>EP-2: When the app is relaunched from the home screen, the app must restore the app state as closely as possible to the previous state.</b>                                                                                                                                                                                                                        |
|                       | Irrelevant notifications    | Media,Messaging,POI, Navigation,IOT,Weather                       | <b>IN-1: The app must display notifications only when relevant to the driver's needs.</b>                                                                                                                                                                                                                                                                             |
|                       |                             | Video,Games,Browsers                                              | IN-2: The app must not post any heads up notifications.                                                                                                                                                                                                                                                                                                               |
|                       | Visual or Text Ads          | Media,Messaging,POI, Navigation,IOT,Weather                       | <b>AD-1: The app must not display text-based advertising other than the advertiser's name or the product name.</b>                                                                                                                                                                                                                                                    |
|                       | Notification Ads            | Media,Messaging,POI, Navigation,IOT,Weather                       | NA-1: The app must not present advertisements through notifications.                                                                                                                                                                                                                                                                                                  |
|                       | Messaging Functionality     | Messaging                                                         | MF-1: The app must receive incoming messages.                                                                                                                                                                                                                                                                                                                         |
|                       |                             |                                                                   | MF-2: Messages must be properly grouped and displayed in the correct order.                                                                                                                                                                                                                                                                                           |
|                       |                             |                                                                   | MF-5: The app must implement a peer-to-peer messaging service and not notification services, such as those for weather, stocks, and sport scores.                                                                                                                                                                                                                     |
|                       | Navigation Functionality    | Navigation                                                        | NF-2: The app draws only map content on the surface of the navigation templates. Text-based turn-by-turn directions, lane guidance, and estimated arrival time must be displayed on the relevant components of the navigation template. Additional information relevant to the drive, speed limit, road obstructions, etc., can be drawn on the safe area of the map. |
|                       | POI Functionality           | POI                                                               | PF-1: The app must provide meaningful functionality relevant to driving.                                                                                                                                                                                                                                                                                              |
|                       | Weather Functionality       | Weather                                                           | WE-1: App must include weather related content, which must be relevant to the user’s current location or a user specified location.                                                                                                                                                                                                                                   |
|                       |                             |                                                                   | WE-2: Weather information on map tiles must be readable and may not include complex legends. Apps may include a maximum of three legends. Apps with multiple legends may have a maximum of three colors, whereas single legend apps may have more than three colors.                                                                                                  |
|                       |                             |                                                                   | WE-5: Weather apps must not show more than five unique weather map annotations in a given view (for example: Temperature markers, wind speed markers, humidity, radar overlay, lightning indicators, road conditions all in the same view).                                                                                                                           |
|                       | Media Controls              | Video                                                             | MC-1: The app integrates with media session. Depending on the content, the app must support either the play/pause or stop playback commands. Additionally, the app must provide title and thumbnail metadata for every media item.                                                                                                                                    |
| Specific Scenarios    | Visual Information on Phone | Media,Navigation,POI, IOT,Weather                                 | <b>VI-1: If the user must go to the phone screen—for example, to act on a permission request—then the app must display a message instructing the user to only look at their phone screen when it’s safe to do so.</b>                                                                                                                                                 |
|                       | Sensitive Data              | Browsers                                                          | SD-1: Before syncing data to the car, browsers that synchronize passwords or payments data must do the following steps: prompt the user to authenticate; notify the user on the car screen their data will be synchronized to the car.                                                                                                                                |
|                       |                             |                                                                   | <b>SD-2: Browsers must not save or allow access to passwords or payment information unless the user can block access to passwords using a profile lock.</b>                                                                                                                                                                                                           |
|                       | Driver Distraction          | Video,Games,Browsers                                              | DD-2: While driving, the app must not be launchable and the app’s UI must not be visible. The app’s audio must stop when the user starts driving and cannot be unpause while driving.                                                                                                                                                                                 |
|                       |                             |                                                                   | DD-3: The app must not be launchable or usable while driving and must not play any audio. Note: If your app plays audio, it must meet the DD-2 requirement.                                                                                                                                                                                                           |
|                       | Voice Commands              | Media,Navigation                                                  | VC-1: The app must support Google Assistant voice commands.                                                                                                                                                                                                                                                                                                           |
|                       | Payments                    | Navigation,POI, IOT,Weather                                       | PA-1: The app must have simple flows if purchases are enabled, using shortcuts such as recent or favorite purchases. The app must not allow any of the following: Setup of payment methods; Multiple items to be selected for purchase; Commitment to recurring payments, such as subscriptions.                                                                      |
|                       | Media Autoplay              | Media                                                             | <b>MA-1: Media app must not autoplay on startup or without user initiated action to select the app or app media.</b>                                                                                                                                                                                                                                                  |
|                       | Messaging Functionality     | Messaging                                                         | MF-3: The user can reply to a message.                                                                                                                                                                                                                                                                                                                                |
|                       |                             |                                                                   | MF-4: The app must use short-form messaging app design patterns. Traditional long-form messaging apps, such as apps for email, are not permitted.                                                                                                                                                                                                                     |
|                       | Navigation Functionality    | Navigation                                                        | NF-5: The app must not provide turn-by-turn notifications, voice guidance, or cluster information when another navigation app is providing turn-by-turn instructions.                                                                                                                                                                                                 |
| Visual Effects        | Contrast                    | Media,Messaging,POI, Navigation,IOT,Weather                       | <b>VD-1: The app must provide icons and colors that meet Android Auto’s contrast requirements.</b>                                                                                                                                                                                                                                                                    |
|                       |                             |                                                                   | <b>UX-1: Touch targets are at least 64dp.</b>                                                                                                                                                                                                                                                                                                                         |
|                       | User Experience             | Video,Games,Browsers                                              | <b>UX-2: Touch targets are at least 24dp apart from each other and 24dp away from screen edges.</b>                                                                                                                                                                                                                                                                   |
|                       |                             |                                                                   | <b>UX-3: Font sizes should be at least 24sp.</b>                                                                                                                                                                                                                                                                                                                      |